Lionel Messi’s Shot Data Visualization

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Figure 1: Word Cloud Generator

Figure 1: Word Cloud Figure 1 highlights the key shot keywords from Lionel Messi's career, with word size reflecting frequency across all 2,632 shots. Dominant terms like "Normal," "Left Foot," and "Open Play" immediately capture his core scoring identity Secondary terms such as "Free Kick," "Penalty," "Half Volley," and "Head" round out his versatility, while rarer techniques like "Diving Header," "Backheel," and "Overhead Kick" hint at the moments of brilliance scattered throughout his 20-year career.

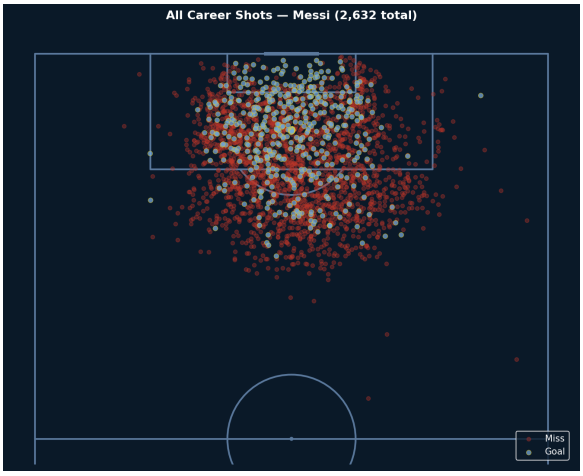

Figure 2: All Career Shots

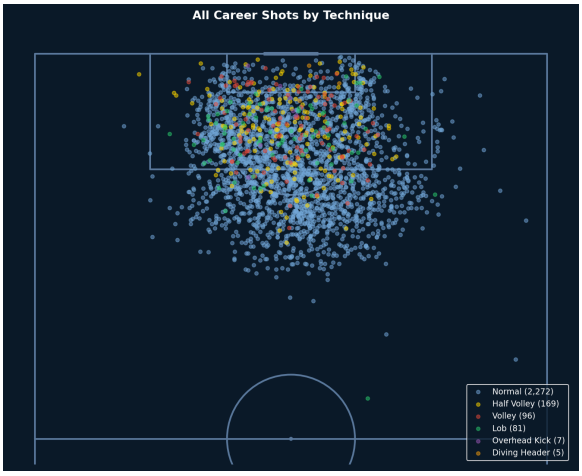

Figure 3: All Shots by technique

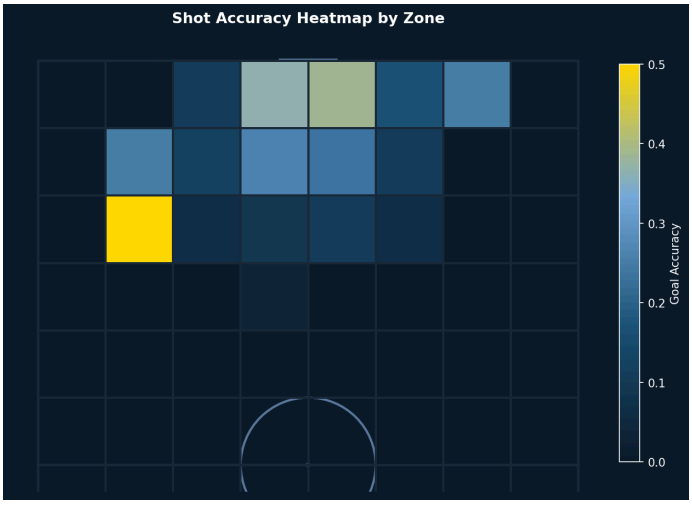

Figure 4: Shot Accuracy Heatmap

Figure 2: All Career Shots Figure 2 plots all 2,632 of Messi's career shots across Barcelona, PSG, and international play, with blue dots representing goals and red representing misses. The dense clustering inside the penalty box highlights his preference for high-percentage positions, while the scattered points beyond the arc reflect his occasional long-range attempts.

Figure 3: All Career Shots by Technique Figure 3 categorizes each shot by technique, revealing that the vast majority (2,272) were standard normal strikes, consistent with Messi's controlled, grounded shooting style. Secondary techniques including half volleys (169), volleys (96), and lobs (81) appear more sparsely distributed across the pitch, typically from wider or deeper positions.

Figure 4: Shot Accuracy Heatmap by Zone Figure 4 visualizes shot accuracy across zones of the attacking half, with gold indicating the highest conversion rates. The standout zone is the left side of the six-yard box, where Messi converts at nearly 50%. Accuracy drops sharply beyond the penalty area, consistent with patterns seen across elite forwards.

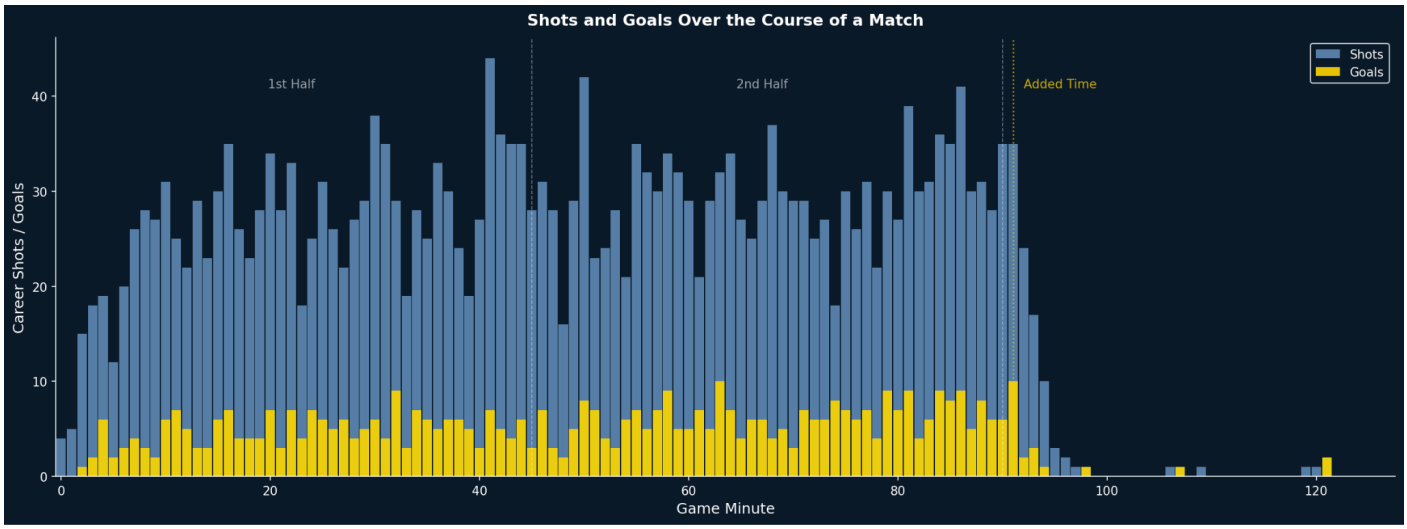

Figure 5: Shots and Goals Over the Course of a Match

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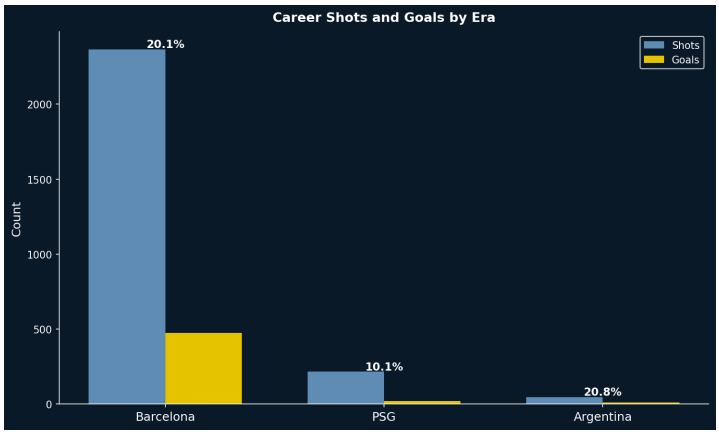

Figure 6: Career Shots and Goals by Era

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of his Barcelona mark. Reflecting his well-documented struggles during his two seasons in Paris.

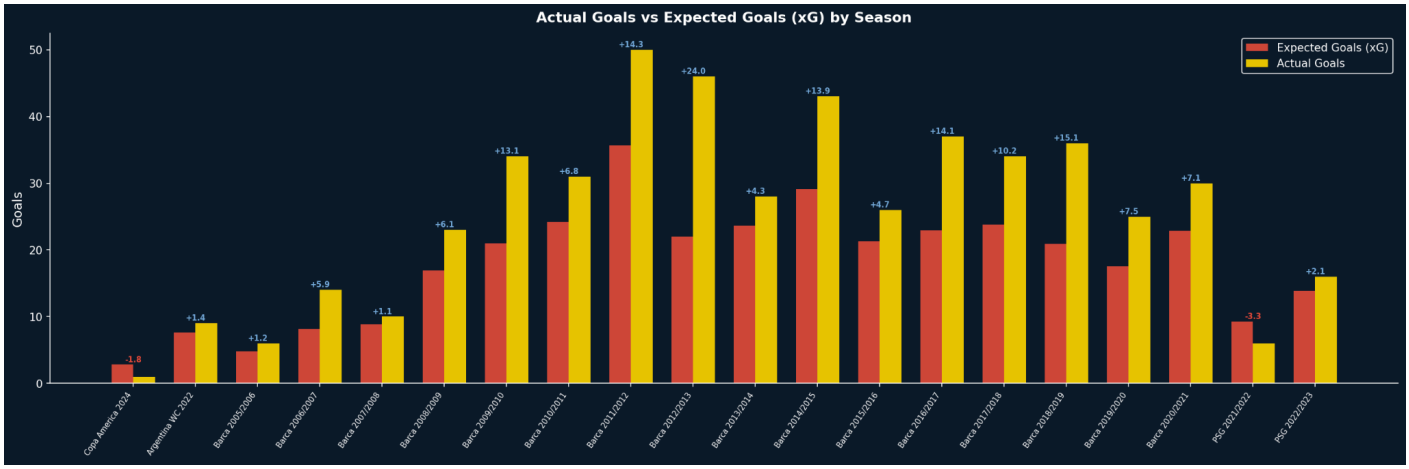

Figure 7: Actual Goals vs Expected Goals by Season

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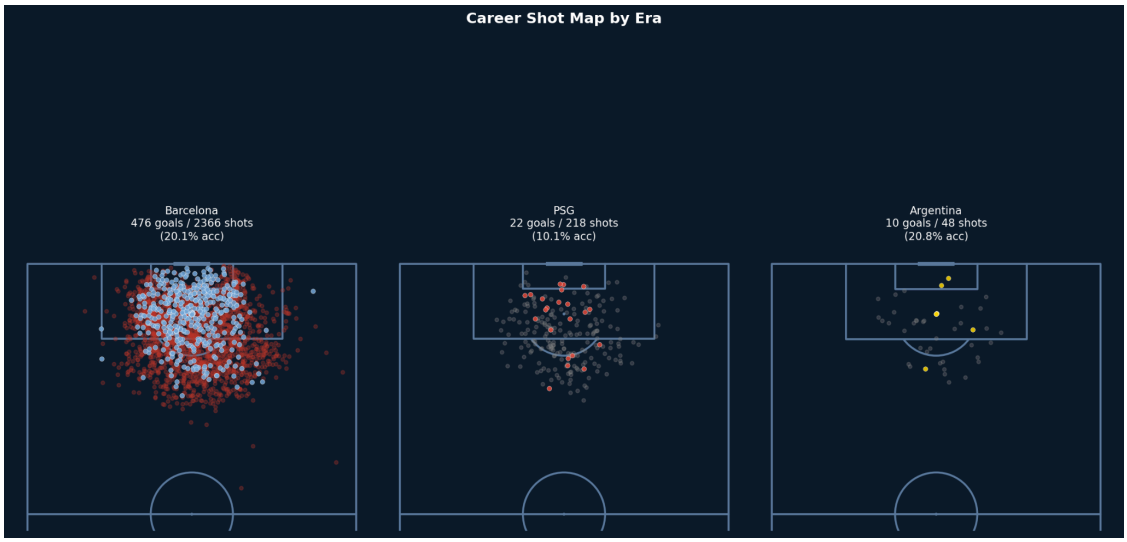

Figure 8: Career Shotmap by Era

Figure 8: Career Shots and Goals by Era Figure 8 breaks down Messi's total shots and goals across each era of his career, with accuracy percentages annotated above each pair of bars. Barcelona dominates in volume with over 2,000 shots and a 20.1% conversion rate, while Argentina matches that accuracy at 20.8% despite a much smaller sample. PSG stands out as the clear outlier, with a 10.1% accuracy rate which is less than half of his Barcelona mark, reflecting his well-documented struggles during his two seasons in Paris.

Importance: With this project, there were some other visualizations that I created but deemed not useful to display the greatness of Lionel Messi. One of them being the teams that he scored the most goals against, I chose to remove this because it doesn't really add anything to the relevance of the visualizations. I feel as the visuals that I used truly display the greatness of Messi on the pitch, with using visualizations of a soccer pitch in a scatter plot, bar charts, and heat maps show where he can do his most damage to the teams that he is playing against.

Dataset: With the help of Claude, I was able to find this open source soccer dataset that takes players goals and puts them in xy coordinates on a soccer pitch, along with how the goal happened and many other different points to what eventually leads to a goal. The photos, code, and additional info is in the github repo that I created for this assignment.

Github Link: https://github.com/cmarsalko/LionelMessiVisualization